

PROVED: 179 OCCURS IN THE FACTORIAL-GCD SEQUENCE OF HARMONIC NUMERATORS

XIAOJIAN ZENG

ABSTRACT. We prove that 179 occurs in the sequence obtained by taking the greatest common divisor of the reduced numerator of the n th harmonic number with $n!$. More precisely, exact integer arithmetic gives $h(31862) = 179$, and the single value $n = 31862$ therefore closes the original existential quantifier. The certificate constructs the harmonic number over an integer common denominator, reduces it exactly, and applies the Euclidean algorithm to the reduced numerator and $31862!$. Four arithmetically distinct implementations reproduce the result, and a fail-closed suite rejects all 34 classes of malformed or wrong-endpoint inputs. No floating-point inference or external theorem is needed for the decisive certificate. We do not claim that 31862 is the least witness, nor do we resolve the wider conjecture that every odd prime occurs.

1. INTRODUCTION

Write the n th harmonic number in lowest terms as

$$H_n = \sum_{k=1}^n \frac{1}{k} = \frac{U_n}{V_n}, \quad U_n, V_n \in \mathbb{Z}_{>0}, \quad \gcd(U_n, V_n) = 1,$$

and put $h(n) = \gcd(U_n, n!)$. The OEIS entry A093818 defines this sequence and records the broader assertion that every odd prime occurs as a conjecture [2]. Its linked table covers $1 \leq n \leq 29524$; among the displayed distinct nonunit values, the primes run through 173 and then jump to 227, so 179 is absent from that published range.

Prime divisibility in harmonic numerators has a substantial adjacent literature. In particular, the sets of indices at which a fixed prime divides the numerator are studied in [1], while Shiu's denominator analysis includes the relevant p -integral block structure near $p(p-1)$ [3]. These results explain why an index of the form $p(p-1)$ is natural, but they do not determine the complete factorial gcd at $p = 179$.

The contribution here is the exact finite identity

$$\gcd(U_{31862}, 31862!) = 179, \quad 31862 = 179 \cdot 178.$$

The identity is enough: the problem asks whether a witness exists, not for the least such witness and not for a proof of the wider all-odd-primes conjecture. The calculation uses arbitrary-precision integers only. We first give

Date: 2026-09-04.

a self-contained structural explanation for the chosen index, then prove the existential statement from a definition-direct certificate, and finally describe four cross-checks and a fail-closed input audit.

A targeted source search through 2026-09-04 found no prior statement of this exact endpoint in the recorded OEIS, arXiv, publisher, and web-index searches. That is a dated, bounded search report, not a claim of absolute world novelty.

2. DEFINITIONS AND THE STRUCTURAL INDEX

For a prime p , let $\mathbb{Z}_{(p)}$ denote the localization of $\mathbb{Z}$ at p : its elements are rational numbers that admit a denominator not divisible by p . The following elementary lemma motivates the certificate index. Its proof is included so that the decisive argument has no external theorem dependency.

Lemma 2.1. *If $p \geq 5$ is prime and $N = p(p-1)$, then*

$$H_N \in p\mathbb{Z}_{(p)}.$$

Consequently, p divides the reduced numerator U_N and does not divide the reduced denominator V_N .

Proof. First we show that $H_{p-1} \in p^2\mathbb{Z}_{(p)}$. Pairing r with $p-r$ gives, in $\mathbb{Z}_{(p)}/p^2\mathbb{Z}_{(p)}$,

$$\frac{1}{r} + \frac{1}{p-r} = \frac{p}{r(p-r)} \equiv -\frac{p}{r^2} \pmod{p^2}.$$

Inversion permutes the nonzero residue classes modulo p , so

$$\sum_{r=1}^{p-1} \frac{1}{r^2} \equiv \sum_{r=1}^{p-1} r^2 = \frac{p(p-1)(2p-1)}{6} \equiv 0 \pmod{p}.$$

The terms indexed by r and $p-r$ agree modulo p ; hence the full sum is twice the half-sum. Since 2 is invertible modulo p , the half-sum also vanishes modulo p . The paired display therefore yields $H_{p-1} \equiv 0 \pmod{p^2}$.

Now split H_N into terms with indices divisible and not divisible by p :

$$H_N = \frac{1}{p}H_{p-1} + \sum_{a=0}^{p-2} \sum_{r=1}^{p-1} \frac{1}{ap+r}.$$

The first term lies in $p\mathbb{Z}_{(p)}$. For each a , the inner sum is in $\mathbb{Z}_{(p)}$ and reduces modulo p to

$$\sum_{r=1}^{p-1} \frac{1}{r} \equiv \sum_{r=1}^{p-1} r = \frac{p(p-1)}{2} \equiv 0 \pmod{p},$$

again because inversion permutes the nonzero residues. Every term in the decomposition is thus in $p\mathbb{Z}_{(p)}$, proving the claim. Reduction to lowest terms then makes the denominator prime to p and the numerator divisible by p . $\square$

For $p = 179$, Lemma 2.1 proves that 179 divides U_{31862} . It does not rule out other prime factors shared by U_{31862} and 31862!; that remaining finite assertion is exactly what the certificate checks.

3. MAIN RESULT

Theorem 3.1. $h(31862) = 179$, and hence $\exists n \in \mathbb{Z}_{\geq 1} h(n) = 179$.

Plain-language endpoint. At $n = 31862$, the reduced numerator U_n of the harmonic number satisfies $\gcd(U_n, n!) = 179$; therefore a positive-integer witness exists.

Proof. Set $n = 31862$ and define the positive integers

$$L = \text{lcm}(1, 2, \dots, n), \quad S = \sum_{k=1}^n \frac{L}{k}, \quad d = \gcd(S, L).$$

Every k in the displayed range divides L , so S is an integer and $H_n = S/L$. Dividing numerator and denominator by d gives a positive coprime pair

$$U = \frac{S}{d}, \quad V = \frac{L}{d}, \quad H_n = \frac{U}{V}.$$

By uniqueness of the positive lowest-terms representation, $U = U_n$ and $V = V_n$.

The exact certificate in Section 4 constructs L and S , performs these integer divisions, and applies the Euclidean algorithm to U and $n!$. The resulting integer is

$$\gcd\left(\frac{S}{\gcd(S, L)}, 31862!\right) = 179.$$

Thus $h(31862) = \gcd(U_{31862}, 31862!) = 179$. Finally, $31862 \in \mathbb{Z}_{\geq 1}$, so this one verified value supplies the witness required by the existential statement. $\square$

4. CERTIFICATE OR COMPUTATIONAL VERIFICATION: EXACT CERTIFICATE AND INDEPENDENT RECOMPUTATIONS

4.1. Definition-direct certificate. The serialized certificate contains only the schema version, $n = 31862$, $p = 179$, the expected value 179, and an exact text binding to the claimed endpoint. It does not contain a stored numerator, denominator, gcd, residue, or other discovery output that a checker could trust instead of recomputing.

The core of the LCM certificate is the following finite integer procedure.

```
n = 31862
L = lcm(1, 2, ..., n)
S = sum(L // k for k = 1, 2, ..., n)
d = gcd(S, L)
U, V = S // d, L // d
assert gcd(U, V) == 1
assert S * V == U * L
assert gcd(U, factorial(n)) == 179
```

The implementation constructs L as the product, over primes $q \leq n$, of the largest power of q not exceeding n . This is exactly $\text{lcm}(1, \dots, n)$. All quotients in the sum are checked integer quotients, reduction uses an exact gcd, and the final assertion is the definition of $h(n)$. No decimal approximation, probabilistic primality claim, or numerical tolerance enters the procedure.

The common reported invariants were

quantity	exact value
n	31862
$h(n)$	179
$\text{bitlength}(U_n)$	45869
$\text{bitlength}(V_n)$	45866
$U_n \bmod 179^2$	$15036 = 84 \cdot 179$
$V_n \bmod 179$	165.

The residue information independently records that the numerator has 179-adic valuation one and the denominator is not divisible by 179. The stronger fact excluding every additional factorial factor is the exact final Euclidean gcd, not the residue table.

4.2. Four arithmetic routes. The certificate was recomputed by four implementations with different fraction constructions:

route	construction	result
LCM sum	Construct $L = \text{lcm}(1, \dots, n)$, sum L/k , and reduce once.	PASS
Reduced recurrence	Start from $0/1$ and replace u/v by $(ku + v)/(kv)$, reducing at every k .	PASS
Balanced tree	Recursively add exact fractions on disjoint index intervals and reduce at each merge.	PASS
Factorial denominator	Use $n!$ itself as a common denominator, sum $n!/k$, and reduce once.	PASS

Each route includes both endpoints $k = 1$ and $k = n$, produces a positive lowest-terms fraction, and then computes the defining gcd with $n!$. All four returned $h(31862) = 179$ and the exact invariants displayed above. The balanced-tree and factorial-denominator checkers use no imported project code; the latter was written in a fresh audit context specifically from the frozen definition. The four routes therefore provide cross-checks of the indexing, common-denominator construction, reduction, and final gcd.

4.3. Fail-closed input audit. Correct arithmetic on a wrongly interpreted endpoint would not certify the theorem. Accordingly, every verifier first enforces the same five-field schema and exact endpoint binding. A separate adversarial suite ran a clean baseline and 34 damaged-input classes against all four implementations. The classes were:

group	examples	count
Syntax and encoding	truncation, nonobject roots, trailing data, invalid UTF-8	6
Keys and schema	missing or extra fields; wrong, Boolean, or floating schema	5
Index domain and type	off-by-one, zero, negative, huge, string, Boolean, float, exponent, NaN	9
Target values	wrong p , wrong expected gcd, Boolean expected gcd	3
Claim binding	empty, nonstring, contradictory, space/NUL mutations, surrogate escape	6
Duplicate members	plain or escaped duplicates in both value orders	5
		34

All four clean baselines passed. Every one of the 34×4 damaged verifier runs was rejected, including duplicate JSON member names after Unicode-escape decoding and contradictory claim text whose other fields were valid. The parsers perform these checks before the expensive harmonic-number reconstruction. This audit does not replace the arithmetic proof; it shows that the certificate interface fails closed rather than silently certifying a different statement.

5. REPRODUCIBILITY

The decisive input is `evidence/witness-31862.json`. The four checker sources are:

- `verification/verify_lcm_sum.py`;
- `verification/verify_recurrence.py`;
- `audit/verify_independent_no_import.py`; and
- `audit/referee_verify_no_import.py`.

From the project root, the reproducible commands are

```
python3 verification/verify_lcm_sum.py \
    evidence/witness-31862.json
python3 verification/verify_recurrence.py \
    evidence/witness-31862.json
printf '%s\n' evidence/witness-31862.json | \
    python3 -I -S audit/verify_independent_no_import.py
printf '%s\n' evidence/witness-31862.json | \
    python3 -I -S audit/referee_verify_no_import.py
python3 audit/referee_fail_closed.py
```

The first four commands reconstruct the harmonic fraction and exact gcd; the last runs the four baselines and the 34-class fail-closed matrix. The two no-import commands use Python’s isolated mode with site initialization disabled. The computation is deterministic and uses no random seed.

6. SCOPE AND LIMITATIONS

The result is deliberately narrow.

- (1) It proves the existence of a witness by exhibiting $n = 31862$. The published OEIS table stops at 29524, but the interval from 29525 to 31861 has not been ruled out here; no least-witness claim is made.
- (2) It does not prove the OEIS conjecture that every odd prime occurs. Only the value 179 in the frozen existential problem is resolved.
- (3) Lemma 2.1 explains the factor 179 at $179(179-1)$, but it does not by itself exclude additional prime factors of $h(31862)$. That exclusion comes from the finite exact gcd computation.
- (4) The prior-art statement is limited to the sources, interfaces, queries, and 2026-09-04 access boundary recorded for this project. Failure to locate a prior result is not an absolute novelty theorem.
- (5) No proof assistant was used. The trust basis is transparent exact integer arithmetic, four distinct reconstructions, strict input binding, and a fresh independent computational audit; this should not be confused with external peer review.

COMPUTATIONAL AND AI DISCLOSURE

AI-assisted research tools were used to help explore candidate indices, prepare and review exact-arithmetic code, formulate adversarial input tests, organize the bounded literature search, and draft the manuscript. The mathematical conclusion was not promoted on model confidence: it was bound to the frozen definition, recomputed by four deterministic integer algorithms, subjected to 34 fail-closed attack classes, and accepted by a fresh independent audit of the exact endpoint. No proof assistant was used. The author remains responsible for the statement, computations, exposition, and final claims.

REFERENCES

1. Leonardo Carofiglio, Luigi De Filpo, and Alessandro Gambini, *p-adic valuation of harmonic sums and their connections with wolstenholme primes*, Indian Journal of Pure and Applied Mathematics **55** (2024), 555–566.
2. Vladeta Jovovic, *A093818: Greatest common divisor of the numerator of the n-th harmonic number and n!*, The On-Line Encyclopedia of Integer Sequences, Including the linked b-file through $n = 29524$; accessed 2026-09-04.
3. Peter Shiu, *The denominators of harmonic numbers (revised)*, 2024, Version 2, revised 30 July 2024.

PBC SCHOOL OF FINANCE, TSINGHUA UNIVERSITY, 100190
 Email address: xiaojianzeng@foxmail.com